

Mental Health and Psychosocial Outcomes Among Individuals With Mayer-Rokitansky-Küster-Hauser (MRKH) Syndrome: A Scoping Review

Running title: Mental health in MRKH syndrome: a scoping review

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Abstract

Background: Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome is a rare congenital condition, affecting approximately 1 in 4,500–5,000 female births, characterized by uterovaginal agenesis in individuals with a 46,XX karyotype and functional ovaries. Typically diagnosed during adolescence, MRKH is associated with significant psychosocial burden, yet mental health outcomes and coping mechanisms remain insufficiently synthesized, limiting their integration into clinical care and public health planning. This scoping review mapped the mental health outcomes, coping mechanisms, and healthcare system gaps reported among individuals with MRKH.

Methods: Following Arksey and O'Malley's framework and the PRISMA Extension for Scoping Reviews (PRISMA-ScR), we searched PubMed, Scopus, and (via EBSCOhost) MEDLINE, CINAHL, APA PsycInfo, and Women's Studies International for studies published between January 2019 and March 2026. Eligible studies were primary quantitative, qualitative, and mixed-methods research reporting psychological outcomes and/or coping mechanisms in MRKH populations.

Results: Of 516 records identified (324 after deduplication), 34 studies were provisionally included. Psychosexual and relational challenges were the most frequently reported outcomes (n = 24; 71%), followed by reduced quality of life, body image, and self-esteem (n = 18; 53%), broader psychological distress (n = 18; 53%), and depression and anxiety (n = 10; 29%). Coping mechanisms were documented in 14 studies, mostly peer support and self-management, and healthcare system gaps in 9. Studies came mainly from Europe (n = 14) and Asia (n = 14).

Conclusions: MRKH-related psychosocial burden is substantial yet under-integrated into care models. Multidisciplinary, mental health-inclusive care and greater research investment, particularly in underrepresented and low-resource settings, are urgently needed.

Introduction

Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome, also termed Müllerian agenesis or Müllerian aplasia, is a congenital condition characterized by absence of the uterus and upper two-thirds of the vagina in individuals with a 46,XX karyotype, functional ovaries, and typical secondary sexual development. It affects approximately 1 in 4,500–5,000 female births¹; a nationwide Danish registry study estimated a prevalence of 1 in 4,982 live female births.² MRKH is classified as type I (isolated uterovaginal aplasia) or type II (accompanied by extragenital anomalies, most often renal, skeletal, auditory, or cardiac).³ Diagnosis is usually made in adolescence during evaluation for primary amenorrhea,^{1,3} a developmental period that is critical for identity formation, body image, and social relationships.

The diagnosis carries profound implications for fertility, psychosexual development, and self-image. Management commonly includes vaginal dilation as first-line treatment, surgical creation of a neovagina when indicated, and fertility options such as gestational surrogacy and, more recently, uterus transplantation.^{1,3} The American College of Obstetricians and Gynecologists identifies psychosocial counseling as one of the most important components of effective management.¹ Earlier studies documented elevated psychological distress and lower self-esteem among women with MRKH compared with controls,⁴ higher anxiety scores among affected adolescents,⁵ and poorer mental health–related quality of life, higher anxiety, and impaired sexual wellness relative to normative data.⁶ A review of this early literature concluded that creating a neovagina alone does not ensure a good psychological outcome and that psychological support at critical times can be helpful,⁷ while qualitative work described how young women manage intimacy, a sensitivity to difference, and threats to identity after diagnosis.⁸ More recent qualitative studies describe emotional turmoil at diagnosis, challenges to

sexual identity and intimate relationships, the profound impact of infertility,⁹ and reduced self-esteem and sexual wellbeing.¹⁰

Two systematic reviews have since synthesized the psychological, quality-of-life, and sexual outcomes of MRKH.^{11,12} However, to our knowledge, the more recent literature has not been mapped with a public health lens that jointly considers mental health outcomes, the coping mechanisms individuals use, and the healthcare system gaps that shape access to psychosocial support. As a result, psychosocial outcomes remain insufficiently integrated into clinical and public health frameworks for this population.

A scoping review is well suited to mapping a heterogeneous body of quantitative, qualitative, and mixed-methods evidence and identifying gaps for future research and practice.^{13,14} This scoping review therefore addressed the following research question: *What mental health outcomes and coping mechanisms are reported among individuals with MRKH syndrome, and what healthcare system gaps exist?*

Methods

Study Design

This scoping review followed Arksey and O'Malley's five-stage methodological framework¹³—(1) identifying the research question; (2) identifying relevant studies; (3) selecting studies; (4) charting the data; and (5) collating, summarizing, and reporting the results—and is reported in accordance with the PRISMA Extension for Scoping Reviews (PRISMA-ScR; Appendix C),¹⁴ informed by published guidance on the steps of conducting a scoping review.¹⁵ A review protocol was not registered.

Eligibility Criteria

Eligibility was defined using the Population–Concept–Context approach. The *population* was individuals diagnosed with MRKH syndrome; the *concepts* were psychological or mental health outcomes and coping mechanisms; and the *context* was any healthcare or community setting in any country. We included primary quantitative, qualitative, and mixed-methods studies published between January 2019 and March 2026 that reported psychological outcomes and/or coping mechanisms among individuals with MRKH. Studies focused solely on anatomical, surgical, or fertility outcomes were excluded, as were records published before January 2019.

[Confirm any additional criteria applied, e.g., language (English only?), peer-reviewed publications only, and exclusion of reviews, case reports, editorials, and conference abstracts.]

Information Sources and Search Strategy

Searches were run and exported on March 29, 2026, in PubMed, Scopus, and EBSCOhost [confirm the PubMed search was also run on this date]. The EBSCOhost search covered MEDLINE Ultimate, CINAHL Ultimate, APA PsycInfo, and Women's Studies International, and APA PsycInfo was also exported separately (Appendix A). Date limits were not applied identically across interfaces: the Scopus search was limited to 2019–2026, the PubMed export covered 2021–2026 (MEDLINE records from 2019–2020 were captured through the EBSCOhost MEDLINE search), and EBSCOhost results were exported without a date limit. Records published before January 2019 were therefore removed during title and abstract screening. Search strategies combined controlled vocabulary and free-text terms for MRKH (e.g., “Mayer-Rokitansky-Küster-Hauser,” “MRKH,” “Müllerian agenesis,” “vaginal agenesis”) with terms for mental health and coping (e.g., “depression,” “anxiety,” “psychological distress,” “quality of life,” “body image,” “coping”). Records retrieved from each source are reported in

Appendix A. [Verify that the example terms above match the search you ran; recover exact strings from your PubMed, Scopus, and EBSCOhost search histories if saved.]

Study Selection

The original screening records could not be recovered, so study selection was repeated from the original export files. Records from all sources were merged and 192 duplicates were identified by matching DOIs and normalized titles, leaving 324 unique records. Titles and abstracts were screened in two steps: objective criteria (publication year and article type) were applied first, and each remaining record was then read against the eligibility criteria. An AI tool (Claude, Anthropic) proposed a decision and reason for every record; [the first author and a second reviewer independently verified each decision, and disagreements were resolved by discussion — complete once verification is done]. Records that could not be classified from the title and abstract were retained for full-text review. The screening workbook, with a decision and reason for every record, is available as supplementary material.

Data Charting

For each included study, data were charted on author, year, country, study design, sample size, reported mental health and psychosocial outcomes, coping mechanisms, and healthcare system gaps. In this version, charting was based on titles and abstracts; [full-text verification of every charted item is pending]. Outcomes were coded into four domains—depression and anxiety; quality of life, body image, and self-esteem; psychosexual and relational challenges; and broader psychological distress—and coping mechanisms into seven categories.

Synthesis

Findings were collated using a descriptive numerical summary—the number and proportion of included studies reporting each outcome domain—and a qualitative content analysis that grouped findings into outcome domains, coping mechanisms, and healthcare system gaps. Because a single study could report multiple domains, domain counts are not mutually exclusive. Consistent with scoping review methodology, a formal critical appraisal of study quality was not performed.^{13,14} Ethical approval was not required because this review analyzed published data.

Results

Selection of Sources of Evidence

The searches identified 516 records. After removal of 192 duplicates, 324 records were screened and 278 were excluded: published before 2019 (n = 89); case report / surgical case series (n = 75); not primary research (n = 48); surgical/functional outcomes only (n = 38); wrong population / topic (n = 28). Of the 46 records retained, 34 were provisionally included and 12 require full text to decide eligibility (Figure 1).

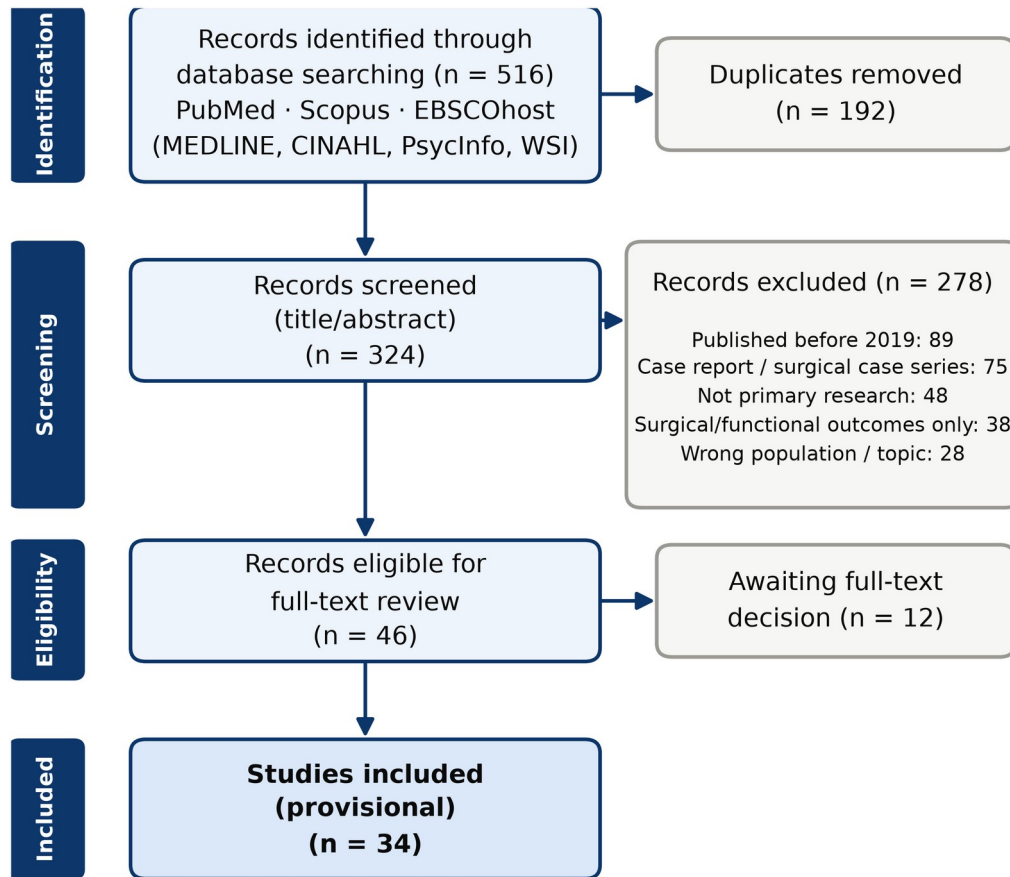


Figure 1. PRISMA-ScR flow diagram of study selection. Counts are provisional pending reviewer verification and full-text assessment.

Characteristics of Included Studies

The 34 included studies were published between 2019 and 2026 and comprised 18 quantitative studies (52.9%), 12 qualitative studies (35.3%), and 3 mixed-methods studies (8.8%); the design of 1 study could not be determined from its abstract (Table 1; Figure 2). Quantitative designs included cross-sectional surveys, case-control comparisons, cohort and pre–post studies, and 1 randomized controlled trial.¹⁶ Studies came from Europe (n = 14) and Asia (n = 14) in equal numbers, followed by North America (n = 3) and Africa (n = 1); 2 studies recruited internationally, and none came from South America or Oceania (Figure 3). China contributed the most studies (n = 6). Sample sizes, where reported, ranged from 7 to 616

participants. Two Chinese reports appear to draw on the same sample of 141 patients.^{17,18}

Characteristics of each study are listed in Appendix B.

Table 1. Study design and geographic distribution of included studies (N = 34)

Characteristic	n	%
Study design		
Quantitative	18	52.9%
Qualitative	12	35.3%
Mixed methods	3	8.8%
Not reported	1	2.9%
Region		
Europe	14	41.2%
Asia	14	41.2%
North America	3	8.8%
Multinational	2	5.9%
Africa	1	2.9%
South America	0	0.0%

Percentages are of all included studies (N = 34) and may not total 100% because of rounding. Classifications are provisional (abstract-level).

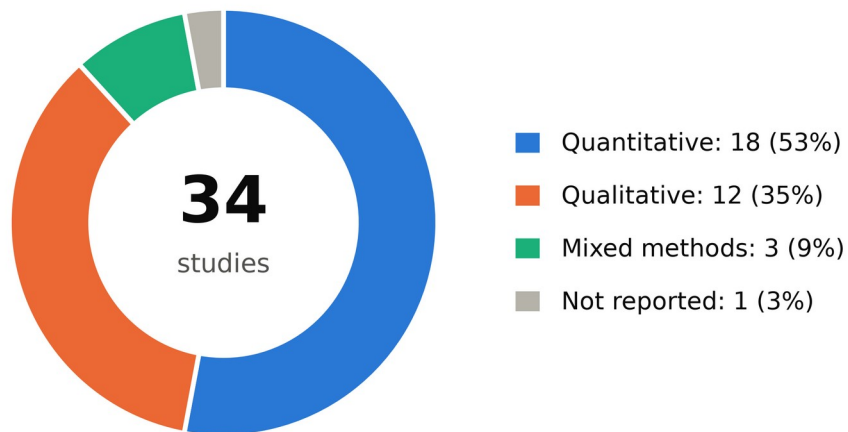


Figure 2. Distribution of included studies by design (N = 34).

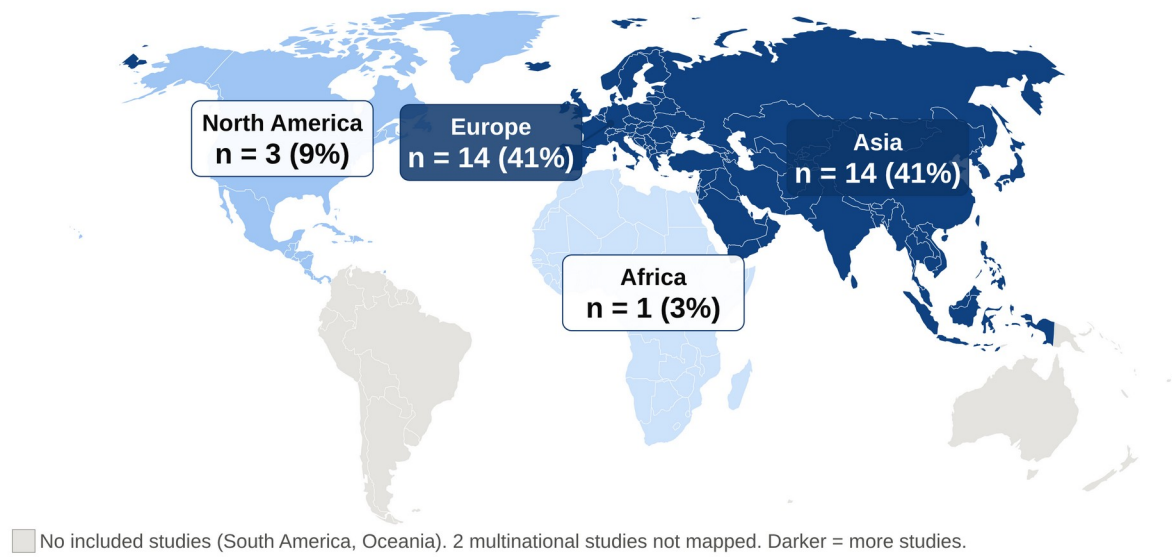


Figure 3. Geographic distribution of included studies by region (N = 34). Darker shading indicates more studies; 2 multinational studies are not mapped.

Mental Health and Psychosocial Outcomes

Psychosexual and relational challenges were the most frequently reported domain (24 of 34 studies; 70.6%), followed by quality of life, body image, and self-esteem (18; 52.9%), broader psychological distress (18; 52.9%), and depression and anxiety (10; 29.4%) (Table 2; Figure 4).

Psychosexual and relational challenges. Compared with controls, women with MRKH reported lower sexual esteem, a more negative genital self-image, and more sexual distress, even after neovagina creation.^{19,20} Qualitative studies described insecurity about the neovagina, low sexual confidence, and anxiety about disclosing the diagnosis to partners.^{10,21–23} One randomized trial found that e-learning psychosexual education improved genital self-image and reduced sexual distress.¹⁶ A French psychodynamic study described how the neovagina shaped sexual identity and body image.²⁴

Quality of life, body image, and self-esteem. Several studies found impaired psychological or mental health–related quality of life,^{25–27} and an international survey of 263

patients reported higher distress and lower self-esteem than the general population.²⁸ Low self-esteem was also reported in a further study.²⁹ Others found similar quality of life across treatment approaches³⁰ or improvements in well-being and body image after treatment.^{31,32}

Broader psychological distress. Qualitative studies described shock, anger, sadness, shame, and secrecy around the diagnosis, and portrayed diagnosis as a turning point that disrupted imagined futures and female identity.^{9,33,34} The diagnostic process itself was experienced as upsetting and potentially traumatizing.³⁵

Depression and anxiety. Where measured with validated scales, findings were mixed. Depressive symptoms were reported in 75.2% of 141 Chinese patients (34.0% moderate to severe),¹⁸ moderate-to-severe anxiety in 24.1%,¹⁷ and depression and anxiety in 32.5% and 37.7% of 77 Malaysian women.³⁶ After vaginoplasty, 45.3% of 53 women reported mild-to-moderate depression and 34.0% mild anxiety.³⁷ A US cohort found anxiety and depressive disorders about twice as common as in male, but not female, referents.³⁸ In contrast, some samples after neovagina creation or awaiting uterus transplantation showed few depressive symptoms.^{39,40}

Table 2. Mental health and psychosocial outcome domains reported in included studies (N = 34)

Outcome domain	Description	Studies, n (%)
Psychosexual and relational challenges	Sexual esteem, sexual distress and wellbeing, intimacy, partner relationships, disclosure	24 (70.6%)
Quality of life, body image and self-esteem	Overall or mental health–related quality of life; body or genital image; global self-esteem	18 (52.9%)
Broader psychological distress	Distress, shame, identity disruption, psychopathology	18 (52.9%)
Depression and anxiety	Depressive and/or anxiety symptoms or diagnoses	10 (29.4%)

Studies could report more than one domain; counts are not mutually exclusive.

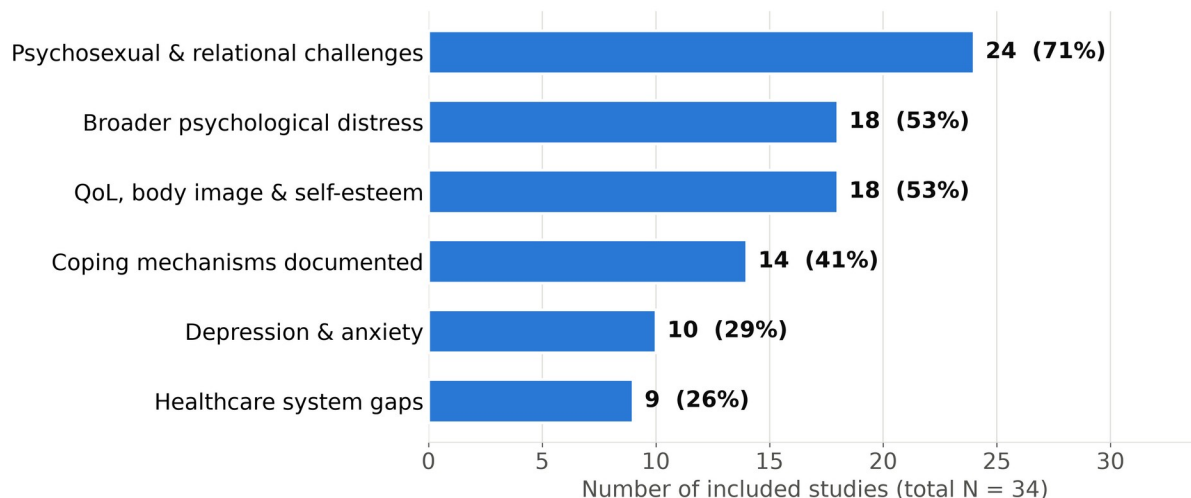


Figure 4. Number of included studies reporting each outcome domain, coping mechanisms, and healthcare system gaps (N = 34). Studies could contribute to more than one category.

Coping Mechanisms

Coping mechanisms were documented in 14 studies (41.2%; Table 3). Peer, family, and social support was most common,^{23,26,28,34,41,42} including MRKH support groups and online communities. Avoidance and concealment—such as pretending to menstruate or hiding the diagnosis—were the most frequently documented maladaptive strategies.^{23,33,41,43} One qualitative study traced a shift from avoidance to empowerment through positive reappraisal and spiritual coping,⁴¹ and illness coherence and positive affect appeared protective for psychological adjustment.²⁸ Structured psychological or psychosexual interventions were evaluated in only 3 studies.^{16,27,44}

Table 3. Coping mechanisms documented in included studies

Coping mechanism	Type	Studies, n
Peer, family & social support	Adaptive	6
Avoidance / concealment (maladaptive)	Maladaptive	4
Psychological counseling/intervention	Adaptive	3
Identity reconstruction / positive reappraisal	Adaptive	3
Spiritual coping	Adaptive	2
Acceptance	Adaptive	2
Self-advocacy	Adaptive	1

A study could document more than one coping mechanism.

Healthcare System Gaps

Healthcare system gaps were reported in 9 studies (26.5%; Table 4). Participants described providers with limited knowledge of MRKH and the need to advocate for themselves,^{33,45} delayed diagnosis and insensitive communication at diagnosis, including stigmatizing language,^{23,35,42} scarce information, psychological support, and fertility counseling,^{42,43,46} and variable counseling around vaginal lengthening treatment.⁴⁷

Table 4. Healthcare system gaps identified and corresponding public health implications

Gap identified	Public health implication
Providers' limited knowledge of MRKH; patients must self-advocate	Train providers in psychosocially informed MRKH care
Delayed diagnosis and insensitive communication at diagnosis	Integrate psychosocial support and mental health screening into diagnostic pathways
Scarce information, psychological support, and fertility counseling	Develop multidisciplinary protocols including psychology and social work; build peer support infrastructure
Variable counseling around vaginal lengthening treatment	Standardize counseling and shared decision-making for treatment
Few studies from Africa and none from South America (research gap)	Expand research investment in underrepresented and low-resource settings

Discussion

Summary of Evidence

This scoping review mapped 34 studies published between 2019 and 2026 on the mental health and psychosocial outcomes of individuals with MRKH syndrome. Psychosexual and relational challenges were the most frequently reported outcome, reported in about seven of every ten studies, while quality of life, body image, self-esteem, and broader psychological distress were each reported in about half. Depression and anxiety were measured less often, and results were mixed. Coping was documented in about four in ten studies and relied largely on peer support and self-management, and about one in four studies described healthcare system gaps.

These findings are consistent with earlier evidence that women with MRKH experience greater psychological distress and lower self-esteem than controls,⁴ higher anxiety in adolescence,⁵ and poorer mental health–related quality of life and sexual wellness.⁶ Prior systematic reviews likewise highlighted poor sexual esteem and genital image.^{11,12} The present review indicates that psychosexual concerns remain central in the recent literature and extends earlier syntheses by mapping coping mechanisms and healthcare system gaps.

Coping and Psychosocial Support

Peer support, reappraisal, and spiritual coping suggest considerable resilience after diagnosis, consistent with earlier qualitative accounts of women managing threats to identity and intimacy.⁸ At the same time, avoidance and concealment were common and may delay help-seeking. Evidence that structured intervention helps is available—a randomized trial of group cognitive-behavioural therapy improved psychological outcomes,⁴⁸ and recent studies of psychological support and psychosexual education reported benefits^{16,44}—yet only a few studies evaluated such support, suggesting it has not been translated into routine care.

Healthcare System Gaps and Equity

Although professional guidance identifies psychosocial counseling as central to MRKH management,¹ participants described providers unfamiliar with the condition, insensitive communication, and scarce psychological and fertility counseling, indicating a gap between recommendations and practice. The evidence base is more geographically diverse than earlier literature—Asia contributed as many studies as Europe—yet only 1 study came from Africa and none from South America, where expectations regarding fertility, marriage, and womanhood and

access to specialized care may differ substantially. Studies from Türkiye, Vietnam, and Japan illustrate how cultural context shapes experiences of diagnosis, disclosure, and infertility.^{9,43,49}

Implications for Practice, Policy, and Research

Five priorities emerge. First, psychosocial support and mental health screening should be integrated into MRKH diagnostic pathways. Second, multidisciplinary care protocols should include psychology and social work alongside gynecology. Third, psychosexual counseling should accompany vaginal lengthening treatment, given the prominence of sexual esteem and genital image concerns. Fourth, providers should be trained in sensitive, non-stigmatizing communication. Fifth, research investment should be expanded in underrepresented and low-resource settings. Future studies should use validated measures, longitudinal designs, and trials of psychosocial interventions delivered within routine care.

Strengths and Limitations

Strengths include an established methodological framework, PRISMA-ScR reporting, a search of biomedical, nursing, psychological, and women's studies databases, and inclusion of quantitative, qualitative, and mixed-methods evidence. The most important limitation is that the original screening records were lost and selection was repeated from the original exports with AI-assisted screening and abstract-level data charting; results are therefore provisional until reviewer verification and full-text assessment of the 46 eligible records are complete. In addition, the review was limited to 2019–2026, date limits were applied inconsistently across database interfaces and enforced at screening, designs and measures were heterogeneous, and some studies may share samples. Consistent with scoping methodology, study quality was not

appraised, and counts reflect how often outcomes were studied, not their prevalence. [Add, if applicable: grey literature not searched.]

Conclusions

MRKH-related psychosocial burden is substantial yet under-integrated into care models. Psychosexual and relational challenges, reduced quality of life and self-esteem, and psychological distress are frequently reported, while coping relies largely on peer support and self-management rather than systematic clinical pathways. Multidisciplinary, mental health–inclusive care and greater research investment are urgently needed, particularly in underrepresented and low-resource settings.

Declarations

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Conflicts of interest: [The authors declare no conflicts of interest. — confirm with all authors.]

Author contributions: [e.g., I.P.S.: conceptualization, search, screening, data charting, analysis, writing—original draft; P.O. and R.A.: supervision, methodology, writing—review and editing.

Edit to reflect actual contributions (CRediT taxonomy).]

Ethics approval: Not applicable; this review analyzed published studies.

Data availability: The data charting form and extracted data are available from the corresponding author on reasonable request.

Use of AI tools: Generative AI (Claude, Anthropic) was used to propose title and abstract screening decisions, chart data from abstracts, draft and edit the manuscript, verify reference details, and prepare figures. The authors reviewed and verified all content and take full responsibility for the work. [Confirm wording against the target journal's AI policy.]

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Appendix A. Search Strategy

Searches were exported on March 29, 2026. Record counts below are taken from the export files and total 516, matching Figure 1. [Paste the exact search string for each database if it can be recovered from your search history.]

[Suggested PubMed/MEDLINE string for comparison. Replace it with the string you actually ran; do not report a search you did not run:]

("Mayer-Rokitansky-Kuster-Hauser"[tiab] OR "Mayer-Rokitansky-Küster-Hauser"[tiab] OR MRKH[tiab] OR Rokitansky[tiab] OR "Mullerian agenesis"[tiab] OR "Mullerian aplasia"[tiab] OR "vaginal agenesis"[tiab] OR "uterovaginal agenesis"[tiab] OR "Mullerian Ducts/abnormalities"[Mesh]) AND (depress[tiab] OR anxiety[tiab] OR "mental health"[tiab] OR psycholog[tiab] OR distress[tiab] OR "quality of life"[tiab] OR "body image"[tiab] OR "self-esteem"[tiab] OR coping[tiab] OR psychosocial[tiab] OR "Mental Health"[Mesh] OR "Adaptation, Psychological"[Mesh] OR "Quality of Life"[Mesh]) AND ("2019/01/01"[dp] : "2026/03/31"[dp])

[Scopus: TITLE-ABS-KEY(...) with the same two concept blocks; PsycINFO and CINAHL: the same keywords plus each database's own subject headings (e.g., APA Thesaurus “Coping Behavior”; CINAHL Headings “Quality of Life”).]

Table A1. Records retrieved by database

Database	Platform	Date searched	Records retrieved
MEDLINE Ultimate; CINAHL Ultimate; Women's Studies International; APA PsycInfo (combined export: 145; 39; 15; 8)	EBSCOhost	Mar 29, 2026	207
APA PsycInfo (separate export)	EBSCOhost	Mar 29, 2026	20
MEDLINE (publications 2021–2026)	PubMed	[Mar 29, 2026?]	146
Scopus (publications 2019–2026)	Scopus	Mar 29, 2026	143
Total			516

Records published before 2019 (100 exported records; 89 after deduplication, all from EBSCOhost) were excluded at title and abstract screening.

Appendix B. Characteristics of Included Studies

Provisional, abstract-level data charting; every entry requires full-text verification. Domains: D, depression/anxiety; Q, quality of life, body image, self-esteem; S, psychosexual and relational; P, broader psychological distress; C, coping documented; H, healthcare system gaps. Reference numbers refer to the main reference list.

Table B1. Characteristics of included studies (N = 34)

Study	Country	Design	n	Domains	Coping
Arsy 2019 ²⁹	Indonesia	Not reported in export (no abstract)	NR	Q	—
Beisert 2022 ²⁰	Poland	Case-control	32 (+32 controls)	D Q S P	—
Blanc 2019 ²⁴	France	Qualitative and quantitative (clinical interviews)	17	Q S P	—
Carroll 2020 ²⁸	Multinational	Cross-sectional online survey	263	Q P C	Peer, family & social support; Identity reconstruction / positive reappraisal
Chen 2020 ¹⁸	China	Cross-sectional (with controls)	141 (+178 controls)	D S	—
Di Mattei 2026 ⁴⁶	Italy	Thematic analysis of online forum	NR	S P H	—
Gilfillan 2024 ³³	United Kingdom	Interpretative phenomenological analysis	13	P C H	Acceptance; Avoidance / concealment (maladaptive)
Güner 2025 ⁹	Türkiye	Phenomenological interviews	10	S P	—
Güner 2026 ⁴¹	Türkiye	Descriptive phenomenological interviews	10	S P C	Avoidance / concealment (maladaptive); Identity reconstruction / positive reappraisal; Spiritual coping; Peer, family & social support
Hatim 2021 ⁴²	Malaysia	Qualitative with quantitative component	12	S P C H	Peer, family & social support
Jensen 2024 ³⁴	Denmark	Semi-structured interviews (life course)	18	S P C H	Peer, family & social support
Jensen 2024 ²¹	Denmark	Semi-structured interviews	18	Q S P	—
Jha 2022 ⁴⁰	India	Case-control	NR	D Q S	—
Kang 2020 ³⁰	China	Cross-sectional	133	Q S	—
Karpel 2025 ³⁹	France	Cross-sectional (uterus transplant candidates)	16 (incl. partners, donors)	D Q S P	—
Khairudin 2026 ³⁶	Malaysia	Cross-sectional	77	D	—
Le 2024 ⁴³	Vietnam	In-depth interviews	20	C H	Avoidance / concealment (maladaptive); Spiritual coping
Lei 2024 ³⁷	China	Longitudinal case series (2 survey rounds)	53	D Q S	—
Lou 2024 ³⁵	Denmark	Interview study	NR	P H	—
Magdoud 2024 ²⁵	Tunisia	Retrospective comparative (with controls)	30	Q S	—
Mao 2024 ³²	China	Retrospective (with controls)	42 (+30 controls)	Q S	—
Marshall 2025 ⁴⁵	United States/Canada	Semi-structured interviews	NR	C H	Self-advocacy
Pennesi 2023 ⁴⁷	Multinational (40 countries)	Cross-sectional mixed-methods survey	616	S P H	—
Rajesh 2026 ¹⁰	Canada	Semi-structured interviews	12	Q S C	Identity reconstruction / positive reappraisal
Rall 2021 ²⁶	Germany	Prospective cohort (pre/post surgery)	82	Q S P C	Peer, family & social support
Sabatucci 2019 ³¹	Italy	Descriptive longitudinal	39	Q S	—
Schäffeler 2022 ²⁷	Germany	Quasi-experimental pre/post	53	D Q S P C	Psychological counseling/intervention
Shao 2022 ⁴⁴	China	Prospective pre/post intervention	30	D C	Psychological counseling/intervention

Song 2020 ¹⁷	China	Cross-sectional (with controls)	141 (+178 controls)	D	—
Sorouri Khorashad 2025 ³⁸	United States	Retrospective cohort	NR	D P	—
Stepanow 2023 ²²	Austria	Semi-structured interviews	10 (+20 controls)	Q S C	Acceptance
Tsitoura 2021 ²³	Greece	Semi-structured interviews	7	S P C H	Avoidance / concealment (maladaptive); Peer, family & social support
Vosoughi 2022 ¹⁶	Iran	Randomized controlled trial	38	Q S C	Psychological counseling/intervention
Weijenborg 2019 ¹⁹	Netherlands	Case-control	54 (+79 controls)	Q S P	—

Appendix C. PRISMA-ScR Checklist

Table C1. PRISMA-ScR checklist (Tricco et al., 2018)

Section / topic	Item	Reported in
Title	1	Title page
Structured summary	2	Abstract
Rationale	3	Introduction
Objectives	4	Introduction (research question)
Protocol and registration	5	Methods: Study Design [complete]
Eligibility criteria	6	Methods: Eligibility Criteria
Information sources	7	Methods: Information Sources; Appendix A
Search	8	Appendix A [complete]
Selection of sources of evidence	9	Methods: Study Selection
Data charting process	10	Methods: Data Charting
Data items	11	Methods: Data Charting
Critical appraisal of individual sources of evidence (optional)	12	Methods: Synthesis (not performed)
Synthesis of results	13	Methods: Synthesis
Selection of sources of evidence	14	Results; Figure 1 [full-text stage pending]
Characteristics of sources of evidence	15	Results; Table 1; Appendix B
Critical appraisal within sources of evidence (optional)	16	Not applicable
Results of individual sources of evidence	17	Appendix B
Synthesis of results	18	Results; Tables 2–4; Figures 2–4
Summary of evidence	19	Discussion: Summary of Evidence
Limitations	20	Discussion: Strengths and Limitations
Conclusions	21	Discussion: Conclusions
Funding	22	Declarations